

Rafael D. De Paz
Independent Researcher
<https://logos.pub>

March 22, 2026

Editorial Board
Journal of Consciousness Studies
Imprint Academic

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

**“Consciousness as Recursive Self-Reference:
Gödelian Limits, Oracle Necessity, and the
Formal Structure of the Observer,”**

for consideration for publication in the *Journal of Consciousness Studies*.

Summary. The paper presents a formal model of consciousness as a recursive self-referential function and proves, via Gödel’s First Incompleteness Theorem, that any sufficiently expressive physical substrate requires a non-algorithmic “Oracle” to resolve undecidable propositions. We identify this Oracle with consciousness and connect the framework to Tononi’s Integrated Information Theory (IIT) and the historical consensus of Planck, Schrödinger, Wheeler, and Bohm.

Contributions.

- A formal consciousness function $C_a = f(M_s, M_l, \varepsilon, E)$ with recursive self-reference as its defining structural property.
- The *Oracle Necessity Theorem*: proof that Gödelian undecidability in physics necessitates a non-algorithmic resolution mechanism.
- A formal correspondence between the Oracle and Tononi’s integrated information $\Phi > 0$.
- A spectral analysis framework linking harmonic coherence (136.1 Hz / 7.83 Hz phase-locking) to the consciousness function.

Suggested Reviewers.

1. **Prof. David Chalmers** (NYU) — Formulator of the Hard Problem of Consciousness.
2. **Prof. Giulio Tononi** (University of Wisconsin–Madison) — Creator of Integrated Information Theory.
3. **Prof. Roger Penrose** (University of Oxford) — Author of the orchestrated ob-

jective reduction hypothesis.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

Respectfully submitted,

Rafael D. De Paz
Independent Researcher
me@rdepaz.com